

1. With **simple interest**, the interest is paid only on the original amount invested, so the same interest I is added every year:

- Find the interest Maria earns each year.
- Write down the value of her investment at the end of years 1, 2 and 3.
- Write down an expression for the value V_n of the investment at the end of n years.
- Find the value of the investment at the end of 12 years.
- Find the number of complete years after which the value of the investment first exceeds \$3000.

Nickname:

2. With **compound interest**, the interest is added to the account and then earns interest itself. If \$ P is invested at $r\%$ per year, compounded k times per year, the value after n years is

$$FV = P \left(1 + \frac{r}{100k} \right)^{kn}.$$

Luca invests \$2000 in Bank A, which pays 3.5% per year compounded **annually**.

- (a) Find the value of Luca's investment at the end of 12 years.

Bank B pays 3.4% per year compounded **monthly**.

- (b) Find the value of \$2000 invested in Bank B at the end of 12 years.
- (c) State which bank gives the greater value after 12 years.
- (d) For Bank A, find the number of complete years after which Luca's investment first becomes at least double its original value.

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